

Install CYW55572x RPi CM4 v6.6.31

Bring up Reference

Guide to Build, flash and configure the platform with v6.6.31 Kernel for CYW5557x support



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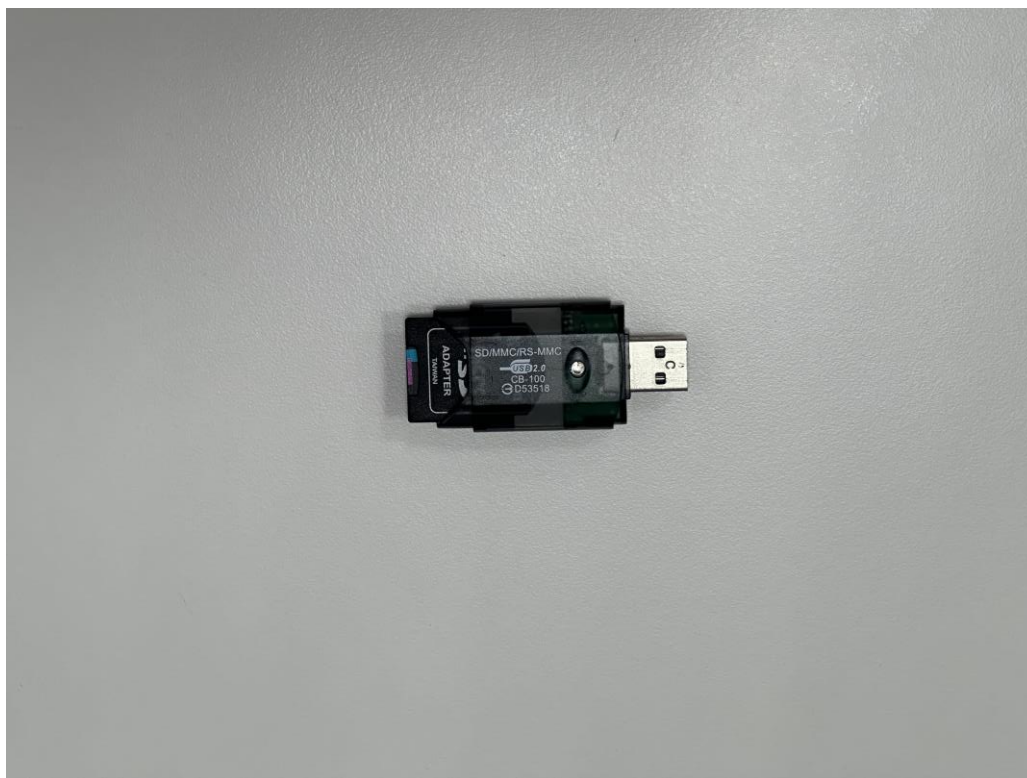
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Set up USDK platform at First Time

1 Set up USDK platform at First Time

- Software Tool: Raspberry Pi Imager
- HW Equipment: For more efficient, recommend using Micro SD reader to plug in desktop to burn image. Reader has usb/type c interface, and it depends on your desktop hardware support.



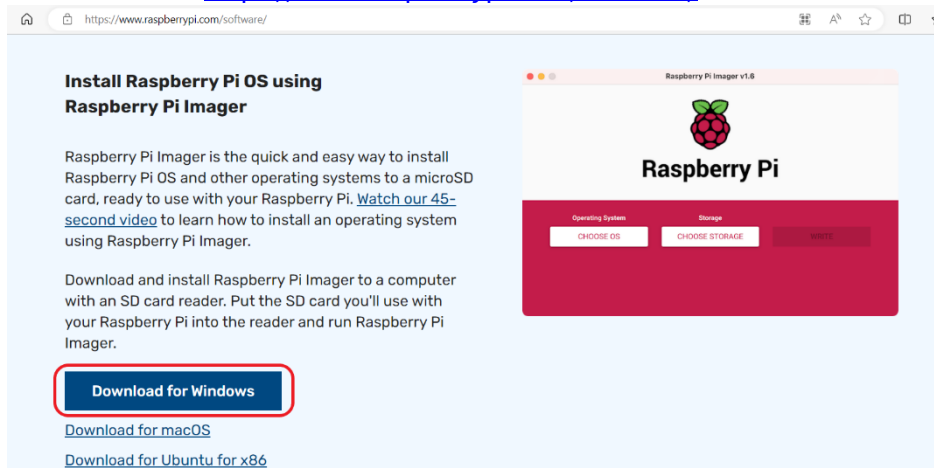
Set up USDK platform at First Time

1.1 Flash USDK Image

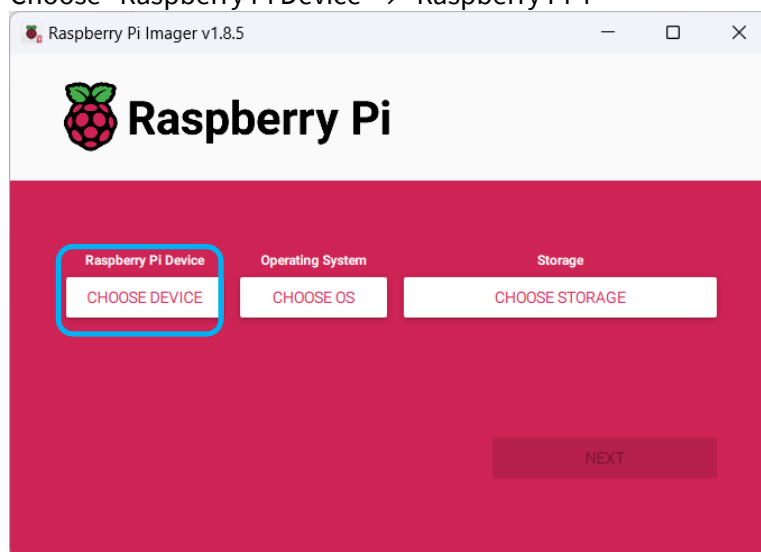
Take the example by Micro SD reader on Windows environment.

- Install Raspberry Pi Imager Choose your environment OS to download tool, such as windows, ios or ubuntu x86. And install the tool on your desktop.

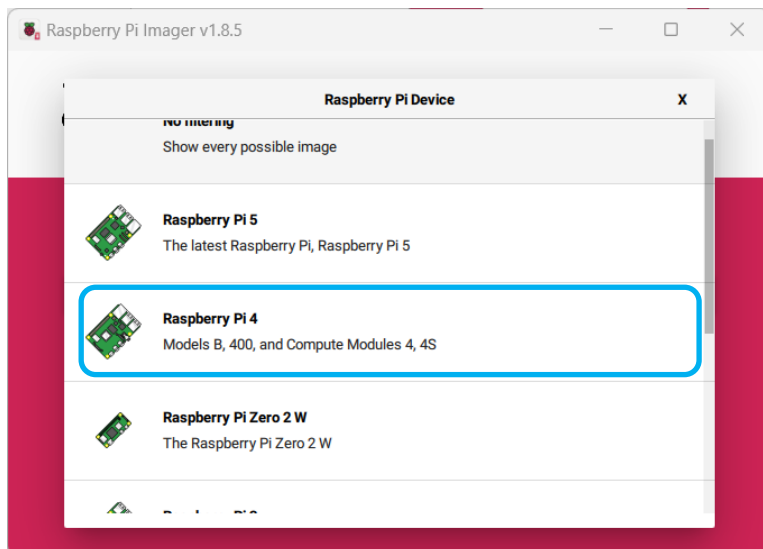
Download Link: <https://www.raspberrypi.com/software/>



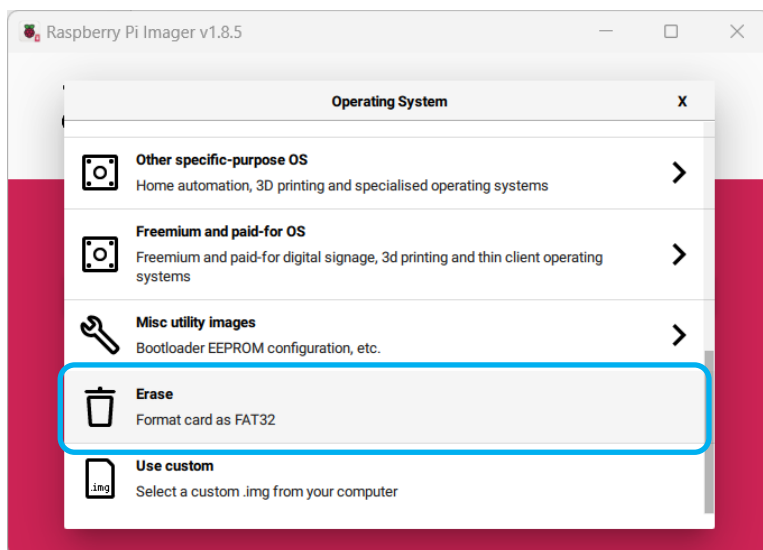
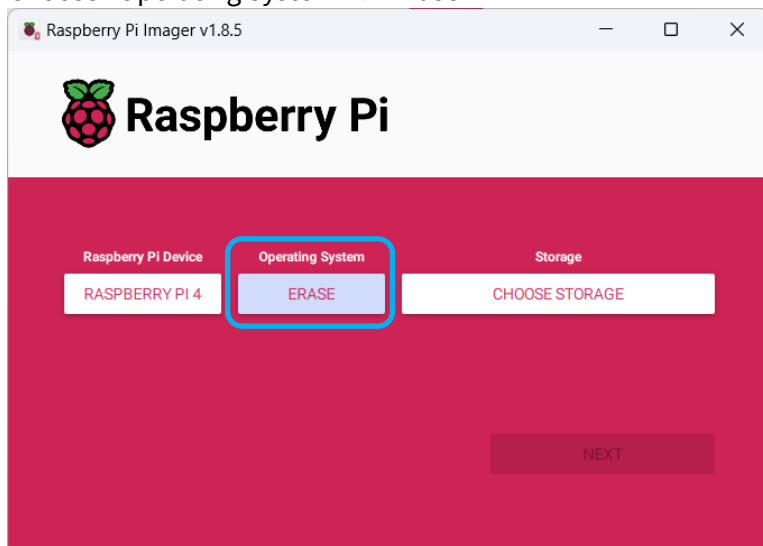
- Download USDK Image from Release Package Link to quick start. Or download from R-Pi official link. But you need to rebuild kernel adding FMac patches for installing modules by manual (Ref: [R-Pi Linux Kernel](#)) and set up environment (section 2).
 - Image Path: rpi-v6.6.31-customer_patch/fmac/rpi_jaculus_img.img.xz
 - Image Login Information:
 - **Root Privileges Account/Password:**
root/12345678
 - **Normal Account/Password:**
utftw/12345678
- Erase the Micro SD Card
 - Choose "Raspberry Pi Device" → "Raspberry Pi 4"



Set up USDK platform at First Time

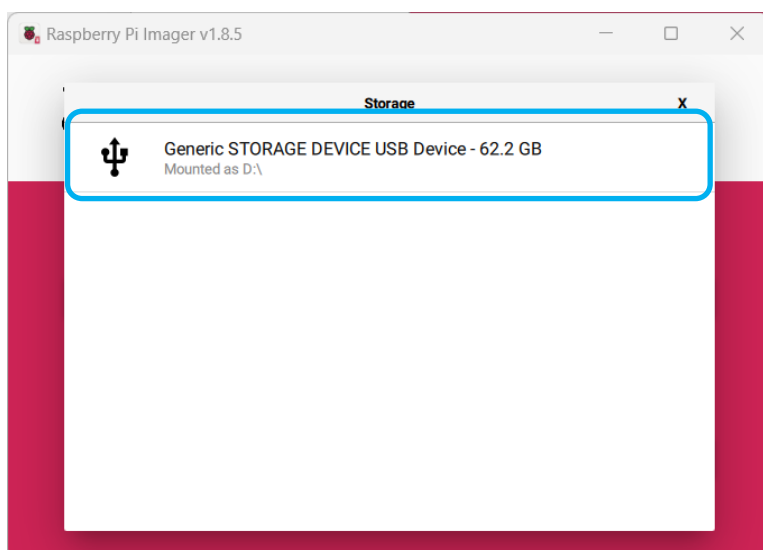
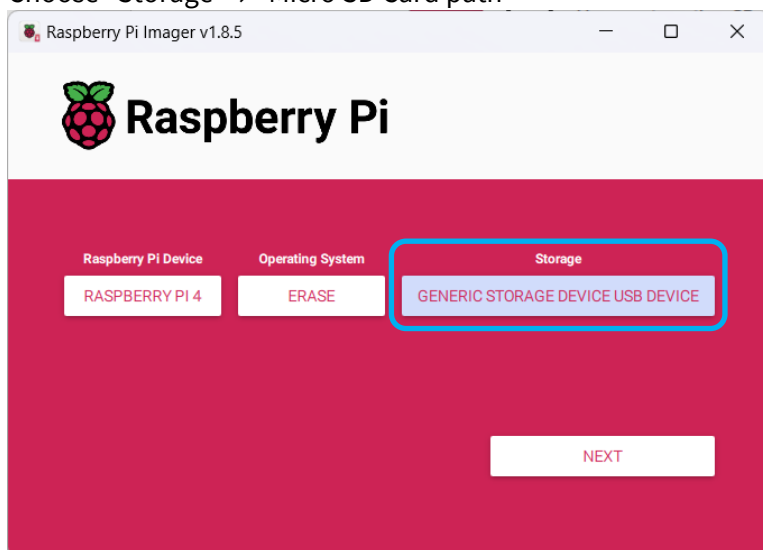


- Choose “Operating System” → “Erase”

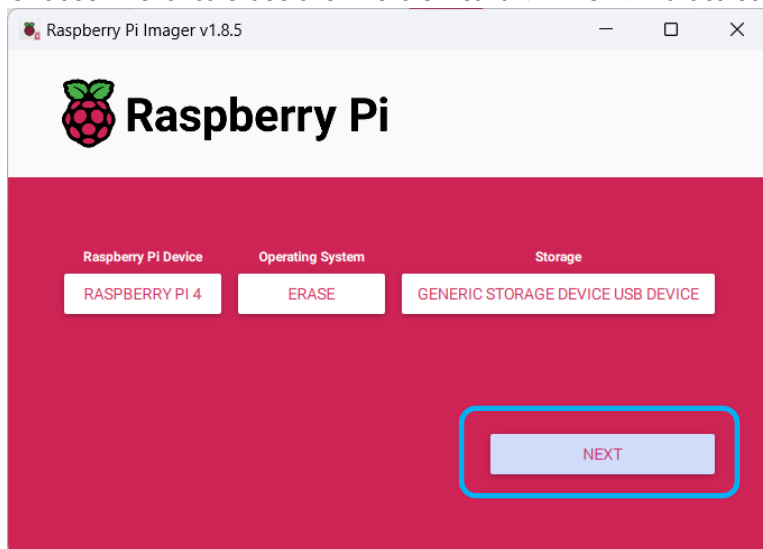


Set up USDK platform at First Time

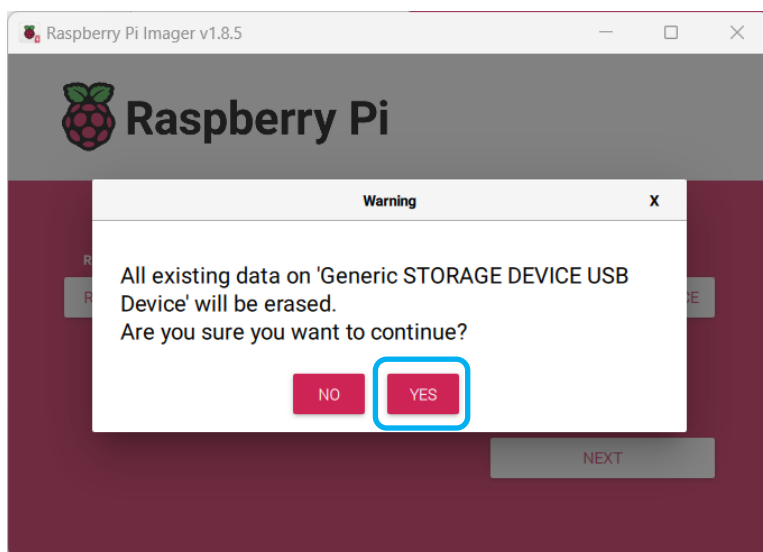
- Choose “Storage” → “Micro SD Card path”



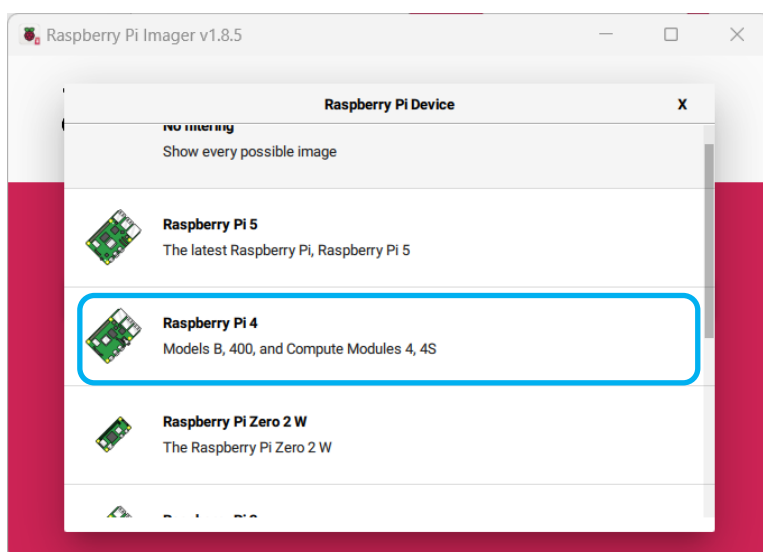
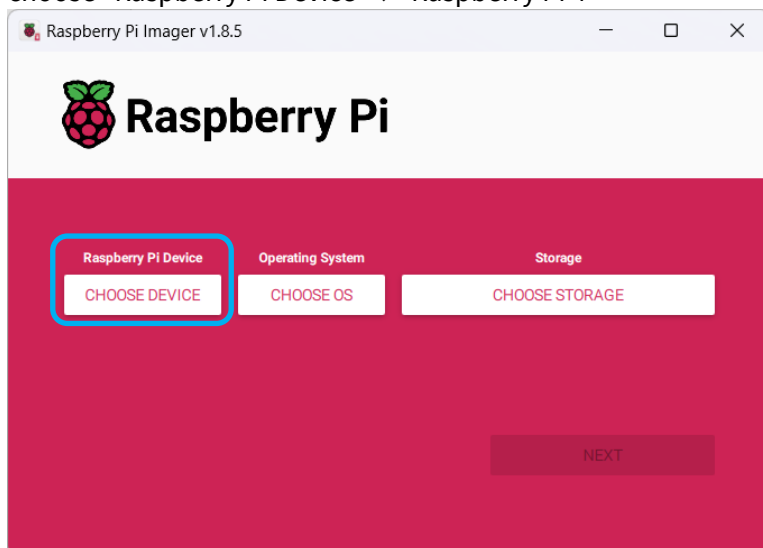
- Choose “Next” to erase the micro SD card → “YES” → wait to complete



Set up USDK platform at First Time

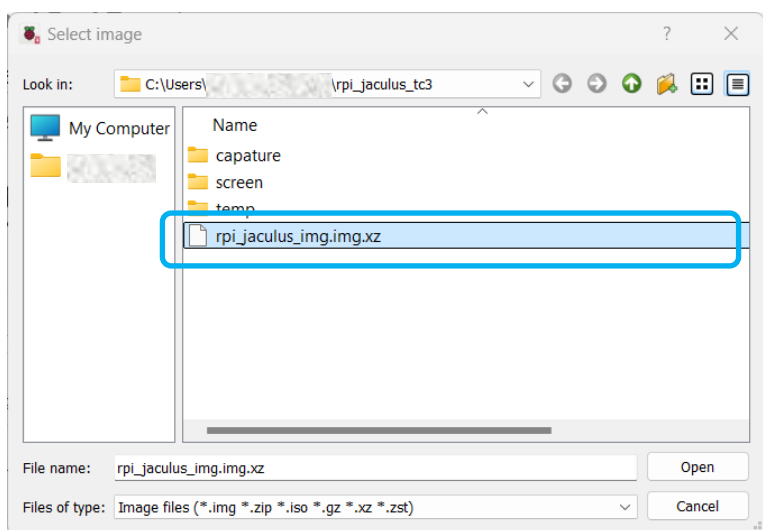
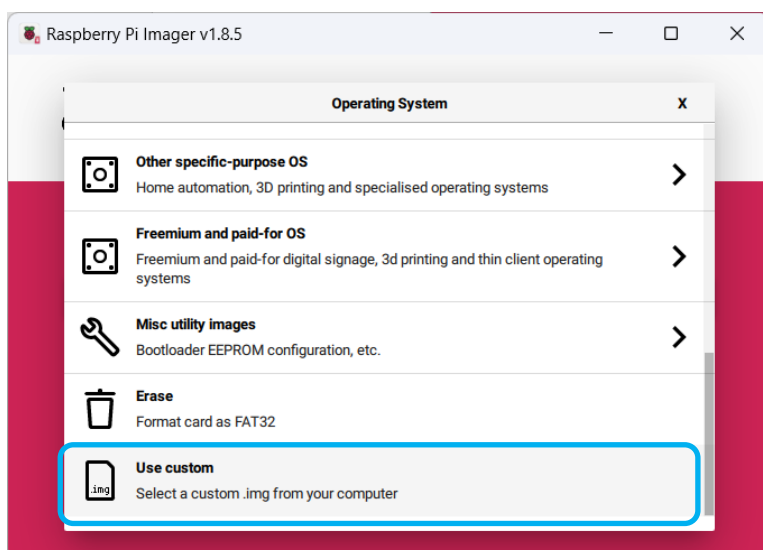
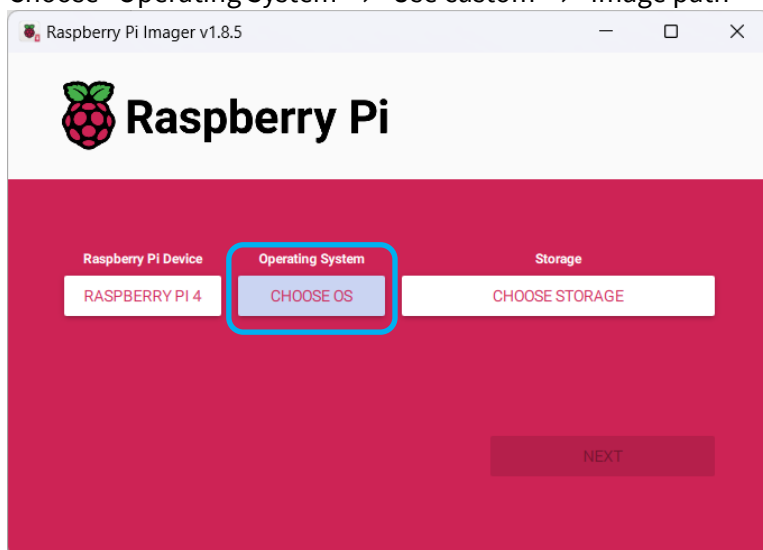


- Flash USDK on RPI Platform
 - choose "Raspberry Pi Device" → "Raspberry Pi 4"



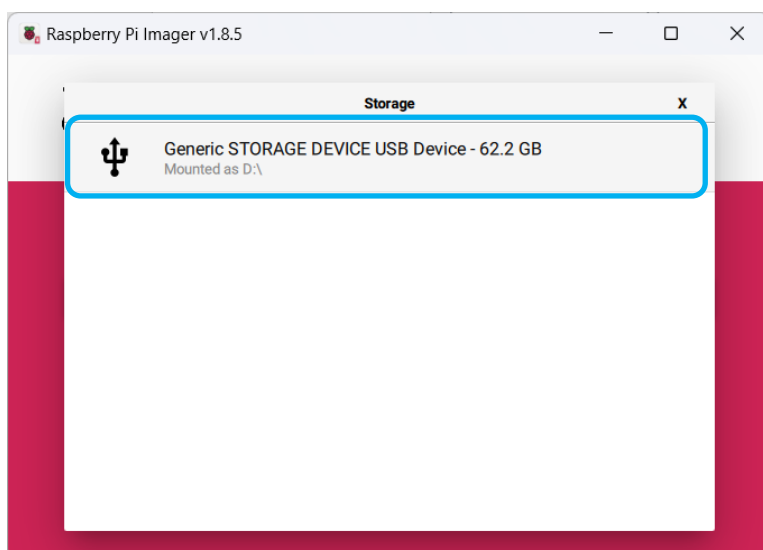
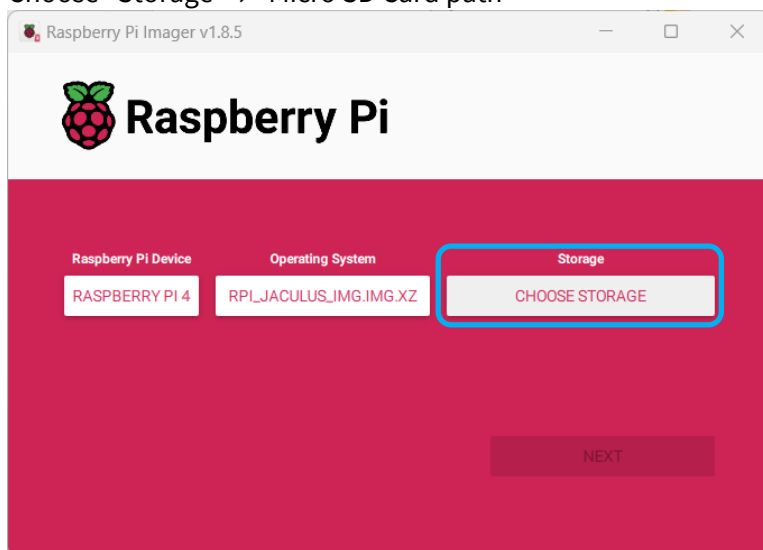
Set up USDK platform at First Time

- Choose “Operating System” → “Use custom” → “Image path”

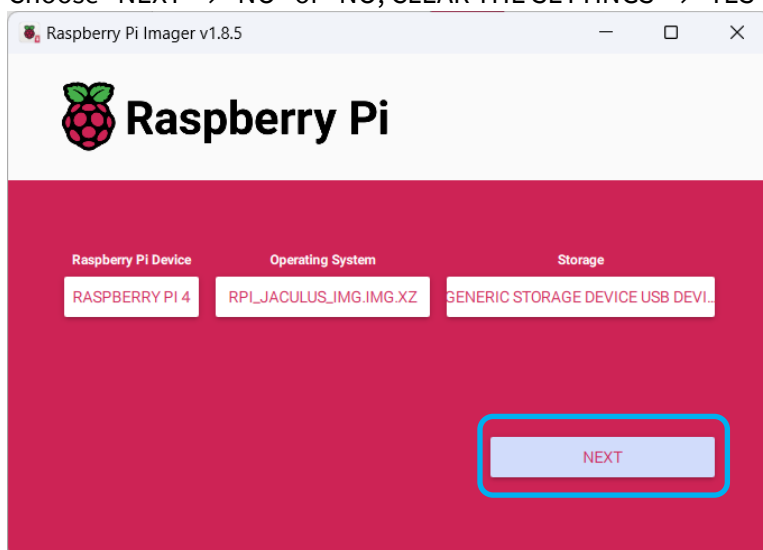


Set up USDK platform at First Time

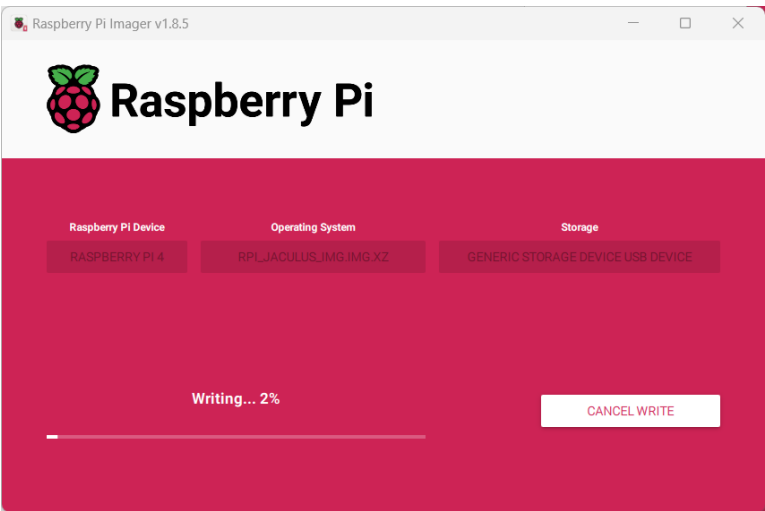
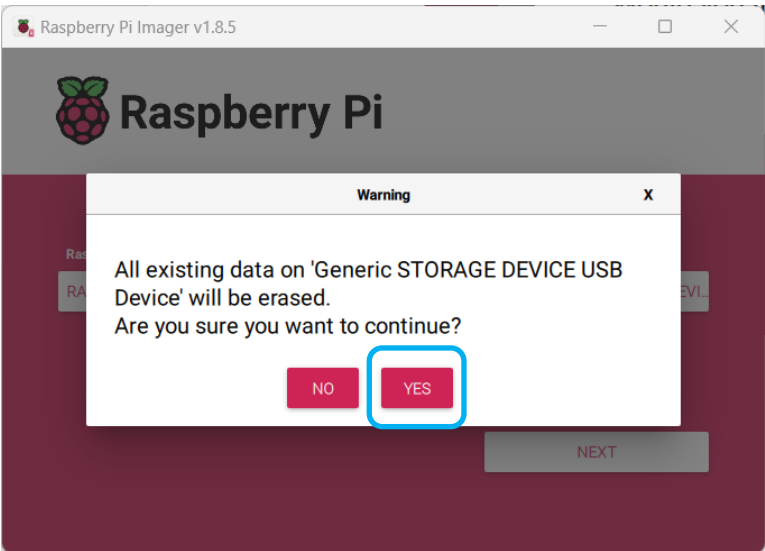
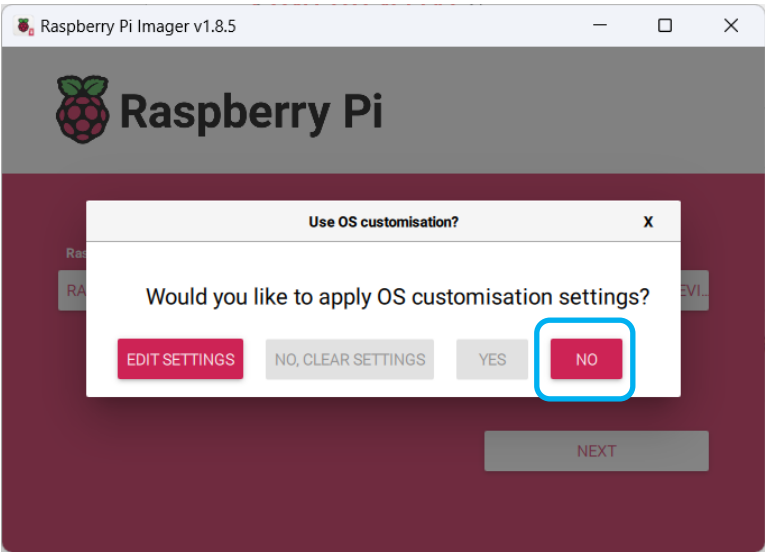
- Choose “Storage” → “Micro SD Card path”



- Choose “NEXT” → “NO” or “NO, CLEAR THE SETTINGS” → “YES” → wait to complete



Set up USDK platform at First Time



Bring up Fmac

2 Bring up Fmac

Upgrade to v6.6 kernel, there is one RPI power sequence issue when use SDIO bus. It needs to press physical buttons “WL_REG_ON” and “BT_REG_ON” after booting up. Pressing purpose is to do unbind MMC driver and then rebind MMC driver. In this section, it will separate into two part to tell PCIE/SDIO bring-up.

2.1 Change nl80211 and openssl libaray

Download libs from release package to copy into linked folder. Some applications such as hostapd and wpa_supplicant are linked to nl80211 and openssl lib so it needs to change these libraries at first time. If this platform burned image from our release, ignore this step.

```
$ sudo cp LIB/* /usr/lib/aarch64-linux-gnu/
```

Bring up Fmac

2.2 Set up Config

- PCIe setting

	PCIE
Add specific bus support under "/boot/firmware/config.txt"	<code>dtoverlay=pcie-32bit-dma</code>
Add specific bus support under "/boot/firmware/cmdline.txt"	<code>pci=pci_bus_perf</code>
Apply bus setting	<code>\$ sudo reboot</code>
Check bus setting	#Look for the following message by "lspci" command. <pre>00:00.0 PCI bridge: Broadcom Inc. and subsidiaries BCM2711 PCIe Bridge (rev 20) 01:00.0 Network controller: Anchor ChipsInc. Device bd31 (rev 01)</pre>

- SDIO setting

	SDIO
Add specific bus support under "/boot/firmware/config.txt"	<code>dtoverlay=sdio</code>
Add specific bus support under "/boot/firmware/cmdline.txt"	<code>none</code>
Apply bus setting	#Notice: It needs to use root privilege. Please switch to root permission. <pre>\$ sudo passwd root \$ su \$ <enter root password></pre>
	#change to root privilege and remove mmc driver <pre>\$ echo fe300000.mmc > /sys/bus/platform/drivers/mmc-bcm2835/unbind 2>/dev/null</pre>
	#reset WL_REG_ON and BT_REG_ON by GPIO toggle disable and enable GPIO2 and GPIO3 <pre>\$ raspi-gpio get 2-3 \$ raspi-gpio set 2 op pn dl \$ raspi-gpio set 3 op pn dl \$ raspi-gpio set 2 op pn dh \$ raspi-gpio set 3 op pn dh</pre>
	#insmod mmc driver <pre>echo fe300000.mmc > /sys/bus/platform/drivers/mmc-bcm2835/bind</pre>
Check bus setting	Look for the following in dmesg log. <pre>\$ dmesg grep SDIO mmc1: new high speed SDIO card at address 0001</pre>

Bring up Fmac

2.3 Put FW and Insert Fmac module

- Change FW

```
$ sudo cp -a <FW/clmb/nvram files path> /lib/firmware/cyress/
```

- Remove Network Manager

Before inserting FMac module, please stop running network manager feature.

```
$ sudo systemctl stop NetworkManager
```

- Before insert fmac modules, we need to copy “brcmfmac-cyw.ko.xz” to the kernel module default folder. Based on v6.6.31 kernel, it supports to vendor firmware API by recognizing chip id. Therefore, system will automatically bring up vendor module when inset brcmfmac kernel module. Ref: [wifi: brcmfmac: add support for Cypress firmware api](#)

```
$ sudo cp brcmfmac-cyw.ko.xz /lib/modules/6.6.31-v8/kernel/drivers/net/wireless/broadcom/brcm80211/brcmfmac/cyw/  
$ sudo insmod ./cfg80211.ko  
$ sudo insmod ./brcmutil.ko  
$ sudo insmod ./brcmfmac.ko
```

2.4 WL tool usage

Please use 64 bits wl tool. Take the example as below. Notice, it needs to change library or it can't excute daemon.

```
$ sudo ./wl_tool_arm64 -h
```

Revision history

Document version	Date of release	Description of changes
1.0.0	09/09/2024	Initializing draft Support to bring up FMac on v6.6.31 kernel.

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